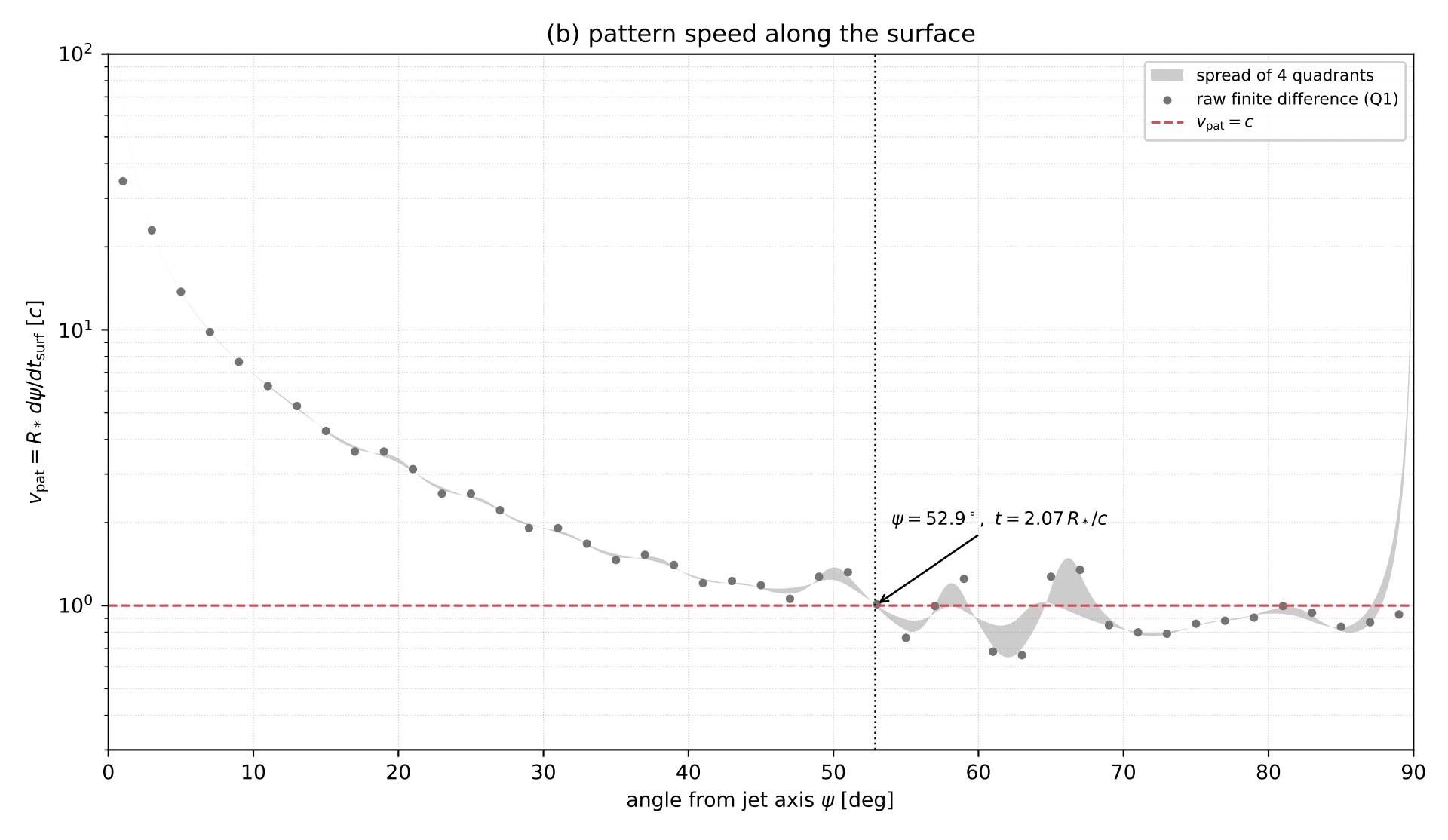
(b) pattern speed along the surface
spread of 4 quadrants
raw finite difference (Q1)
$v_{pat} = c$
$\psi = 52.9^\circ,\ t = 2.07\,R_*/c$
$v_{pat} = R_* \, d\psi/dt_{surf}$ [c]
angle from jet axis ψ [deg]
10^2
10^1
10^0
0
10
20
30
40
50
60
70
80
90